

Original Project Brief

The following pages reproduce the complete assessment brief supplied for the Campus One Digital Academy final project.



ESOFT METRO CAMPUS



Final Project

ESOFT METRO CAMPUS

Shaping Lives, Creating Futures.

Original project brief page 1

Final Project – Assessment Sheet	
Programme	L3 - Diploma in Information Technology
Assessor/Lecturer	
Unit	Final Project
Project Title	
Student's Name	
Student's Registration No	
Submission Date	
Marks Achieved (%)	
Assessor/Lecturer Feedback	

Important Points.

- Carefully check the hand in date and the instructions given with the project. **Late submissions will not be accepted.**
- Ensure that sufficient time is spent to complete the assignment by the due date.
- Do not wait till the last minute to get feedback on the project. Such excuses will not be accepted for late submissions.
- You must be responsible for efficient management of your time.
- If you are unable to hand in your project on time and have valid reasons such as illness, you may apply (in writing) for an extension.
- Failure to achieve at least 40.00% marks will result in a Repeat grade.
- Non-submission of work without valid reasons will lead to an automatic Repeat. You will then be asked to complete an alternative project.
- If you use other people's work or ideas in your assignment, it must be properly referenced, using the HARVARD referencing system, in your text or any bibliography. Otherwise, you'll be found guilty of committing plagiarism.
- If you are caught plagiarising, your grade will be reduced to a Repeat or at worst, you could be excluded from the course.

Project Guidelines

- You need to **create a software** solution for Campus One Digital Academy using **visual studio** as your final project.
- Use **C# Programming language** for your project
- Follow the structure given in this document. Design given forms and apply value additions for it according to your requirement.
- Software **should** contain **minimum two forms**. Additional forms and features will get additional marks. (Login form and one Registration form is compulsory)
- Please follow the given instructions before you start the project.

Instructions

- Design the software system according to the given instructions.
- Use appropriate fonts, colors, and images for Frontend (GUI design) of your software.
- Proposed Software system **must have a database and properly connected** to the system. (Database should be designed using SQL Server Management Studio)
- Database must contain minimum **one Table** (Table column structure provided in Figure X).
- **Test the software** and submit the project on or before the given deadline.
- Create a **Microsoft Word Document** including all the screenshots of forms in your system and code snippets of each form.
- The entire project brief (These nine pages) should be attached as the first section of your project documentation.

Word Processing Rules

- The font size should be 12 point and should be in the style of Time New Roman.
- Set line spacing to 1.15. Justify all paragraphs.
- Ensure that all headings are consistent in terms of size and font style.
- Use the spell check and grammar check function of the word processing application to review the use of language on your assignment.

Proposed Project

Figure 1:- Login

- Above form consist of;
 - One Picture Box
 - One Group Box
 - Three Labels
 - Campus One Digital Academy
 - Username
 - Password
 - Two Text Boxes
 - Username
 - Password
 - Three Buttons
 - Clear
 - Login
 - Exit

Design the above form using these tools and change respective properties using suitable values.

Student Registration - Campus One Digital Academy

[Logout](#)

Campus One Digital Academy

Student Registration

Reg No

Basic Details

First Name

Last Name

Date of Birth

1/ 1/2015

Gender

☐ Male
☐ Female

Contact Details

Address

Email

Mobile Phone

Home Phone

Parent Details

Parent Name

NIC

Contact Number

RegisterUpdate

ClearDelete

[Exit](#)

Figure 2:- Registration Form

- Above form consist of;
 - Four Group Boxes
 - Student Registration
 - Basic Details
 - Contact Details
 - Parent Details
 - Thirteen Labels
 - Campus One Digital Academy
 - Reg No
 - First Name
 - Last Name
 - Date of Birth
 - Gender
 - Address
 - Email
 - Mobile Phone
 - Home Phone
 - Parent Name
 - NIC
 - Contact No
 - Nine Text Boxes
 - First Name
 - Last Name
 - Address (Multiline)
 - Email
 - Mobile Phone
 - Home Phone
 - Parent Name
 - NIC
 - Contact No
 - Four Buttons
 - Register
 - Update
 - Clear
 - Delete
 - Two LinkLabels
 - Logout
 - Exit
 - Two Radio Buttons
 - Male
 - Female
 - One DateTimePicker
 - Date of Birth
 - One ComboBox
 - Reg No

Design the above form using these tools and change respective properties using suitable values.

Project Instructions

Designing

- Design the project using Microsoft Visual Studio. **Designing above two forms is a must**, you can add more forms based on your imagination and those will be considered when marking.
- **Change the properties** of each object with suitable values.

Functional Operations

- When user runs the project, Login form needs to be appeared.

Form I – Login Form (Figure I)

- There is **one user** in the system. User name is “Admin” & Password is “Campusone@123”. When button after applying “Admin” as the username and “Campusone@123” as the Password, the system needs to **hide the login form** and **open the second form** which is Registration form (Figure II). If the login credentials are wrong (Either Username, either Password or both), system needs to generate an error message (Figure III).

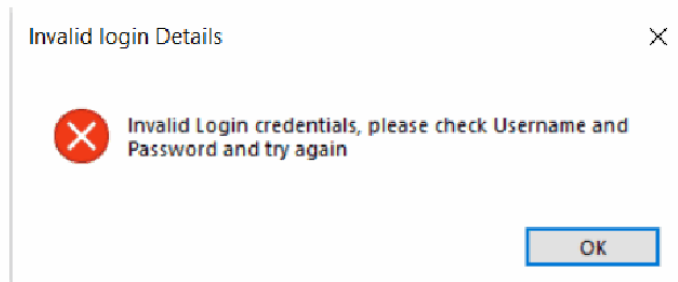


Figure III – Error Message

- You can **increase number of users** for the system. If so, there needs to be a table in the database to track those usernames and passwords as shown in figure IX.
- When user clicks on Clear () button, texts in both username textbox and password textbox needs to be cleared and the username textbox needs to be **focused**.
- When you click on the Exit Button () it needs to appear the following message box.

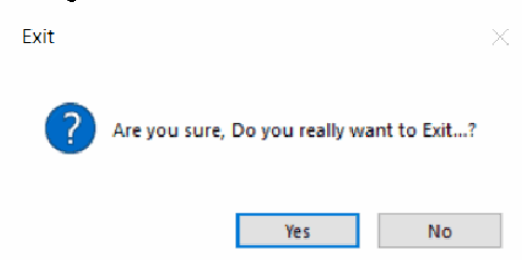


Figure IV – Exit Message

- If you click on Yes button, the entire system needs to be closed and if you click on No, the system stays without a change, but the dialogue box needs to be closed.

Form II – Registration Form (Figure II)

- The Form II (Figure II) is designated form for Student Registrations.
- Create the Database using Microsoft SQL Server Management Studio. Name the database as **Student**.
- Create the **Registration Table**.
- Connect the created Student Database with your C# project.
- In student Registration form, a registration can be done by clicking on Register Button (). Once the user clicks on **Register Button** after providing all the details in related fields in the registration form, the details need to be added in the “**Registration**” Table (Figure X) of “**Student**” Database and needs to appear following message box as well.

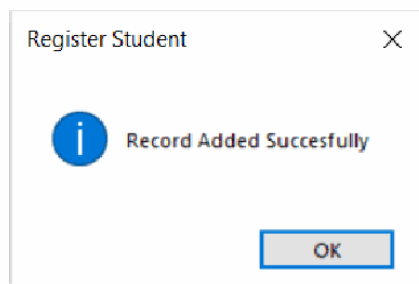


Figure V – Register Message Box

- Give the facility to the user to update records using **Update Button** ().
- Once the update executed, show the following message box.

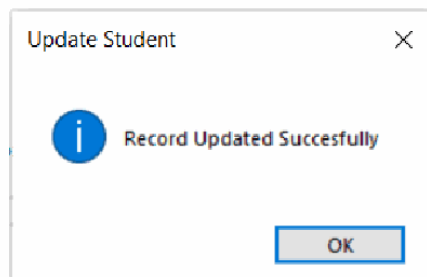


Figure VI – Update Message Box

- Give the facility for the user to Delete a selected record from the Database Table using the **Delete Button** (). Before Execution of Delete function, need to get users confirmation to delete. For that, use following message box or similar one.

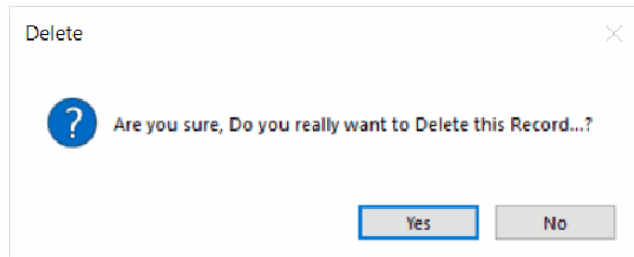


Figure VII – Delete Message Box

- Once the user clicks on Yes from the appearing message box, the details needs to be deleted from the system database based on the Registration Number of the record. After the execution of the Delete, following message box needs to be appeared.

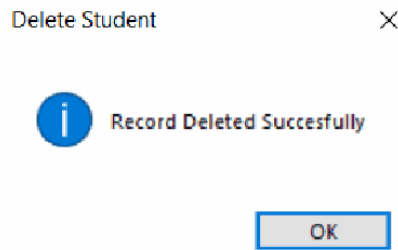


Figure VIII – Delete Confirmation Message Box

- When the user clicks on the **Reg No dropdown list** (**Reg No**), the user can see the registration numbers (Previously registered) appearing in the drop down list. When the user selects a registration number, it needs to work as a **Search** function when user changes the text of this and need to fill all other fields in the form by focusing the Reg No from the Database Table.
- When you click on the Exit Link Label (**Exit**) it needs to appear the Exit message box (Figure IV).
- When user clicks on Logout Link Label (**Logout**), it needs to logout from the system, disappear the Registration form and appear the login form (Figure I).

Database Information

Set Database name as “**Student**” and Registration Table (Figure X) needs to be there inside. When designing the **Registration** Table in the Database, apply **regNo** column as the **primary key** for the table.

	username	password
**	NULL	NULL

Figure IX – Database Table - Logins

regNo	firstName	lastName	dateOfBirth	gender	address	email	mobilePho...	homePhone	parentName	nic	contactNo
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure X – Database Table - Registration

Table Structure

Use the following table to identify data types for each column when you create the “**Registration**” table.

Field Name	Datatype
regNo	Integer
firstName	varchar(50)
lastName	varchar(50)
dateOfBirth	dateTime
gender	varchar(50)
address	varchar(50)
email	varchar(50)
mobilePhone	Integer
homePhone	Integer
parentName	varchar(50)
nic	varchar(50)
contactNo	Integer

Campus One Digital Academy Student Registration Management System

Final Project Report



Student Name: _____

Student Registration Number: _____

Programme: _____

Submission Date: _____

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1 Project Overview

This project implements a Windows desktop system for Campus One Digital Academy. It authenticates an administrator and provides facilities to register, retrieve, update and delete student records stored in Microsoft SQL Server. The completed application also includes a searchable student-records screen, summary totals, selection-based editing and CSV export.

1.1 Project Objectives

- Provide controlled access through the required Login form.
- Store student and parent information in the required Student database.
- Implement reliable Create, Read, Update and Delete operations using parameterized SQL.
- Present a clear interface that follows the supplied form structure.
- Add useful record browsing, searching, reporting and validation features.

2 Requirements and Technology

Area	Implementation
Development environment	Microsoft Visual Studio compatible solution
Language	C#
Framework	.NET 8 Windows Forms
Database	Microsoft SQL Server / SQL Server Express
Database access	ADO.NET with Microsoft.Data.SqlClient 7.0.2
Database name	Student
Main table	Registration

3 System Design

The solution uses three forms. The Login form is the entry point. Successful authentication opens the Registration form, which performs all compulsory student operations. The Student Records form is an additional screen opened from Registration and displays database records in a DataGridView.

Form	Purpose
frmLogin	Authenticates the administrator and controls application entry.
frmRegistration	Registers, retrieves, updates and deletes student records.
frmStudentRecords	Lists, searches, summarizes, exports and selects records for editing.

4 User Interface Implementation

4.1 Login Form

The Login form contains the required PictureBox, Login GroupBox, three labels, two textboxes and three buttons. The password is masked. Clear resets both fields, Login validates the supplied administrator credentials, and Exit requests confirmation.

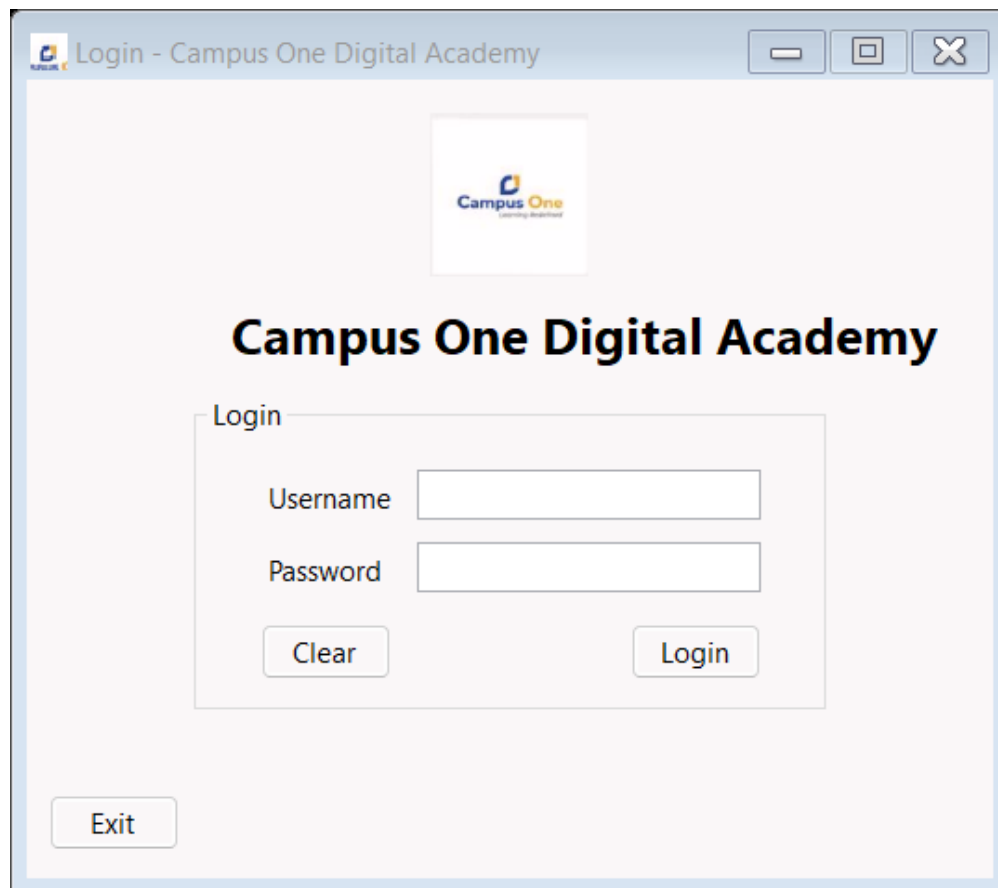
The image shows a Windows-style application window titled "Login - Campus One Digital Academy". Inside the window, there is a logo for "Campus One Learning Accelerated" at the top center. Below the logo, the text "Campus One Digital Academy" is displayed in a large, bold, black font. Underneath this, there is a "Login" label above a light gray rectangular container. Inside this container, there are two labels: "Username" and "Password", each followed by a white text input field. Below the input fields, there are two buttons: "Clear" and "Login". Outside the container, at the bottom left of the window, there is an "Exit" button. The window has standard Windows window controls (minimize, maximize, close) in the top right corner.

Figure 1 Login form

4.2 Student Registration Form

The Registration form contains the four required GroupBoxes and every field specified in the assessment. The enlarged layout separates labels from input controls and keeps Logout, action buttons and Exit visible.

Student Registration - Campus One Digital Academy

[Logout](#)

Campus One Digital Academy

Student Registration

Reg No:

Basic Details

First Name:

Last Name:

Date of Birth:

Gender: ☐ Male ☐ Female

Contact Details

Address:

Email:

Mobile Phone: Home Phone:

Parent Details

Parent Name:

NIC:

Contact No:

Register Update View Students Clear Delete

[Exit](#)

Figure 2 Student Registration form

4.3 Student Records Form

The additional Student Records form displays registered students in a read-only DataGridView. It supports live filtering, refresh, CSV export and returning the selected registration number to the Registration form for editing.

Student Records - Campus One Digital Academy

Registered Students

Search by Reg No, name or NIC

Total Students: 0 Male: 0 Female: 0

Figure 3 Student Records form

5 Database Design

The database is named Student and contains the Registration table. The registration number is the primary key. The application uses the exact field names and SQL Server types supplied in the assessment brief.

Field	SQL type	Purpose
regNo	INT PRIMARY KEY	Unique student registration number
firstName	VARCHAR(50)	Student first name
lastName	VARCHAR(50)	Student last name
dateOfBirth	DATETIME	Student date of birth
gender	VARCHAR(50)	Selected gender
address	VARCHAR(50)	Student address
email	VARCHAR(50)	Student email address
mobilePhone	INT	Student mobile number

homePhone	INT	Student home number
parentName	VARCHAR(50)	Parent or guardian name
nic	VARCHAR(50)	National identity number
contactNo	INT	Parent contact number

6 Core Functionality

6.1 Login

The application compares the entered username and password with Admin and Campusone@123. Correct credentials hide Login and open Registration. Incorrect credentials display an error.

6.2 Register

The Register operation validates every required field, checks for a duplicate registration number and inserts the record with parameters.

6.3 Search and Load

Registration numbers are loaded into the ComboBox. Selecting or entering an existing number retrieves the complete student record.

6.4 Update

Update modifies the matching database row and reports whether a record was found.

6.5 Delete

Delete requires a registration number and confirmation before removing the matching row.

6.6 Clear Logout and Exit

Clear resets fields and focus. Logout returns to Login. Exit closes the application only after confirmation.

7 Additional Features

Feature	Benefit
Student Records DataGridView	Provides a complete, sortable overview of stored students.
Live search	Filters by registration number, first name, last name or NIC while typing.

Dashboard totals	Shows total, male and female counts for the displayed records.
Edit selected record	Returns the selected registration number to the main form.
CSV export	Creates a portable spreadsheet-compatible copy of displayed records.
Window branding	Uses the supplied Campus One logo and title-bar icon.

8 Validation and Error Handling

The Registration form uses ErrorProvider messages beside invalid controls. It validates positive registration numbers, required names and addresses, email format, numeric phone fields, gender selection, NIC entry and future dates. SQL operations use exception handling and display clear database messages.

9 Testing

Test	Expected result	Status
Start application	Login form opens first	Pass
Correct login	Registration form opens	Pass
Incorrect login	Login error appears	Pass
Clear login	Credentials clear and Username receives focus	Pass
Register valid student	Record is inserted and success appears	Pass
Duplicate Reg No	Duplicate warning appears	Pass
Select Reg No	Student details populate	Pass
Update record	Matching row changes	Pass
Delete No	Record remains	Pass
Delete Yes	Record is removed	Pass
Live search	Grid filters while typing	Pass
CSV export	Displayed rows are written to CSV	Pass
Release build	Build completes with zero code warnings or errors	Pass

10 User Guide

- Start SQL Server Express and run DatabaseScript.sql in SQL Server Management Studio if the Student database has not been created.
- Run the application and log in with Username Admin and Password Campusone@123.
- Enter all student, contact and parent information, then select Register.
- Choose a registration number to retrieve a student. Change the details and select Update when required.
- Select Delete and confirm Yes to remove the selected student.
- Select View Students to search, export or choose a record for editing.
- Use Logout to return to Login and Exit to close the program safely.

11 Problems and Solutions

Problem	Solution
Registration table not found	Added an idempotent database script and startup schema check.
Labels overlapped input fields	Enlarged the form and separated label and textbox coordinates.
Logout overlapped the logo	Kept branding in the title bar and repositioned Logout.
Limited record navigation	Added a searchable DataGridView screen with selection-based editing.
Generic validation feedback	Added field-specific ErrorProvider messages and format checks.

12 Conclusion

The completed system satisfies the compulsory Login and Student Registration requirements and connects the required workflow to Microsoft SQL Server through parameterized ADO.NET commands. The additional records screen, search, summary, CSV export and detailed validation improve usability while preserving the required database structure.

13 Code Evidence

13.1 Login Form Authentication

```
if (username == AdminUsername && password == AdminPassword)
{
    Hide();
    frmRegistration registrationForm = new frmRegistration();
    registrationForm.FormClosed += (s, args) => Close();
    registrationForm.Show();
}
else
{
    MessageBox.Show("Invalid Username or Password. Please try again.");
}
```

13.2 Registration Insert

```
string sql = @"INSERT INTO Registration
(regNo, firstName, lastName, dateOfBirth, gender, address, email,
mobilePhone, homePhone, parentName, nic, contactNo)
VALUES
(@regNo, @firstName, @lastName, @dateOfBirth, @gender, @address,
@email, @mobilePhone, @homePhone, @parentName, @nic, @contactNo)";
using SqlCommand command = new SqlCommand(sql, connection);
AddStudentParameters(command);
command.ExecuteNonQuery();
```

13.3 Student Records Search

```
SELECT regNo, firstName, lastName, dateOfBirth, gender, address,  
       email, mobilePhone, homePhone, parentName, nic, contactNo  
FROM Registration  
WHERE @search = " OR CONVERT(VARCHAR(20), regNo) LIKE @pattern  
       OR firstName LIKE @pattern OR lastName LIKE @pattern OR nic LIKE @pattern  
ORDER BY regNo
```

13.4 Database Creation

```
-- SQL Script to create the Student database and required Registration table.  
  
-- Create the Student database  
IF DB_ID(N'Student') IS NULL CREATE DATABASE Student;  
GO  
  
USE Student;  
GO  
  
-- Registration table: stores student registration details  
IF OBJECT_ID(N'dbo.Registration', N'U') IS NULL  
CREATE TABLE Registration (  
    regNo INT PRIMARY KEY,  
    firstName VARCHAR(50) NOT NULL,  
    lastName VARCHAR(50) NOT NULL,  
    dateOfBirth DATETIME NOT NULL,  
    gender VARCHAR(50) NOT NULL,  
    address VARCHAR(50) NOT NULL,  
    email VARCHAR(50) NOT NULL,  
    mobilePhone INT NOT NULL,  
    homePhone INT NOT NULL,  
    parentName VARCHAR(50) NOT NULL,  
    nic VARCHAR(50) NOT NULL,  
    contactNo INT NOT NULL  
);  
GO
```